

Intelligent Contract Analyzer & Summarizer

An LLM + RAG-powered contract intelligence platform that automates first-pass legal review, risk detection, and reporting for law firms and legal teams.

PROJECT OVERVIEW

A production-oriented Legal AI application that automates contract review using Large Language Models, Retrieval-Augmented Generation (RAG), and LangChain. Legal teams upload contracts and receive structured clause extraction, risk scoring, red-flag detection, and an interactive AI chat interface for follow-up review — all packaged into client-ready reports.

Intake & Extraction

- PDF / DOCX upload, batch processing, raw text input
- Payment, termination, confidentiality clauses
- Liability, IP, governing law, dispute resolution

Risk & Analysis

- Automated risk scoring engine
- Red-flag detection with rationale
- Improvement & negotiation recommendations
- Executive summary generation

Reporting & Interaction

- Markdown & PDF report export
- AI chat interface on contract content
- Side-by-side contract comparison
- Contract history dashboard

BEFORE → AFTER

BEFORE • MANUAL PROCESS

- Clause-by-clause manual reading of every contract
- Review quality varies by reviewer experience and fatigue
- Risks surface only if a reviewer happens to catch them
- Contracts sit in a review queue for days
- Ad hoc notes; no searchable record of past reviews
- Re-reading the full document to answer a follow-up question

AFTER • AI-POWERED PLATFORM

- Key clauses extracted and structured automatically in minutes
- Standardized, repeatable risk scoring on every contract
- Red flags and risk score generated automatically, with rationale
- First-pass analysis and report ready the same day
- Structured PDF/Markdown reports plus a contract history dashboard
- Follow-up questions answered instantly via the AI contract chat

TECHNICAL ARCHITECTURE

Backend	FastAPI
AI Orchestration	LangChain (LCEL) — document loaders, RecursiveCharacterTextSplitter, LCEL chains, Pydantic output parsing, conversation memory, RAG
LLM Layer	Groq Llama 3.1 70B (primary) with OpenAI GPT-4o fallback for resilience
Vector Store	Chroma, indexed via Groq Embeddings
Metadata Store	SQLite for contract history and session tracking
Frontend	Streamlit
Document Processing	PyMuPDF, python-docx

AI PIPELINE

Upload → Processing → Cleaning → Chunking → Embeddings → Vector Store → Parallel Chains → Summary + Clauses + Risk → Report → Chat

BUSINESS VALUE

- ▶ Faster, more consistent first-pass contract review
- ▶ Reduced manual effort on high-volume, repetitive review work
- ▶ Improved visibility into legal and contractual risk
- ▶ Professional, client-ready reports for internal and external use

Outcome: Automated, explainable first-pass contract review that gives legal teams structured insight, consistent risk scoring, and decision-ready reports — cutting review turnaround and freeing lawyers to focus on judgment, not extraction.

ENGINEERING CHALLENGES SOLVED

- ▶ Reliable handling of long, dense legal documents
- ▶ Structured, schema-validated LLM outputs via Pydantic
- ▶ Retrieval accuracy tuning for legal terminology
- ▶ Multi-format ingestion with explainable risk output